

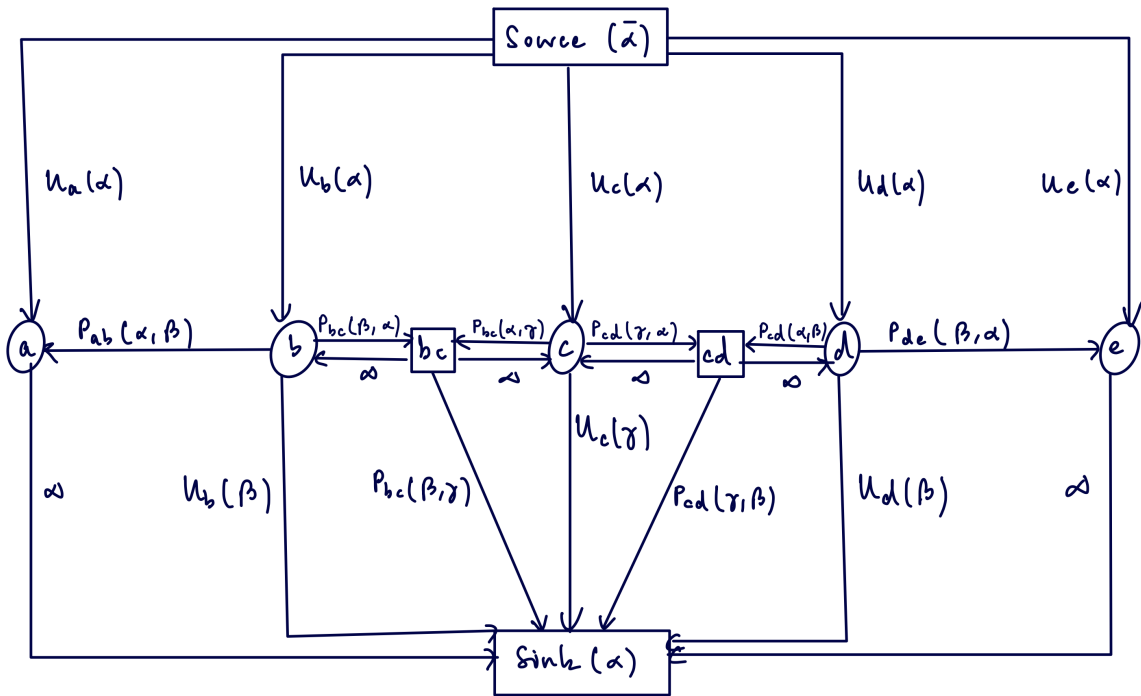
Report: CV Sheet 05

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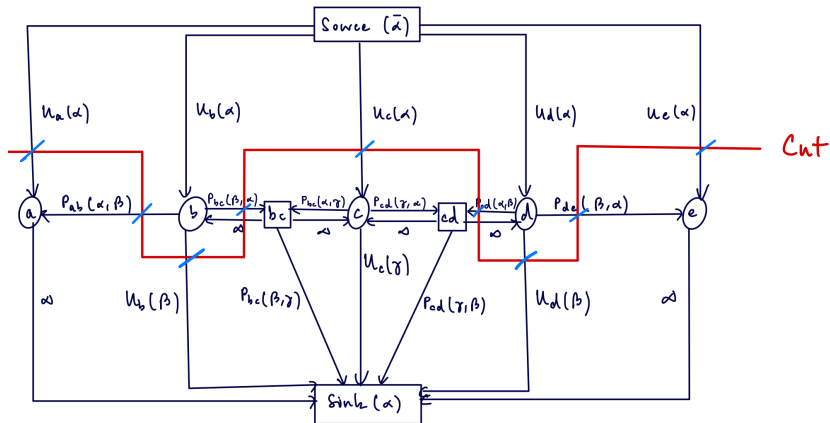
1 Multi-Label MRFs

1.1 Alpha-Expansion Graph Construction



1.2 The Minimum Cut

Show Cut for configuration: $\alpha | \beta | \alpha | \beta | \alpha$



Costs:

$$u_a(\alpha) + P_{ab}(\alpha, \beta) + u_b(\beta) + P_{bc}(\beta, \alpha) + u_c(\alpha) + P_{cd}(\alpha, \beta) \\ + u_d(\beta) + P_{de}(\beta, \alpha) + u_e(\alpha)$$

2 Binary Graph Cut for Interactive Segmentation

2.1 Energy Formulation

We implement a binary Graph Cut model for interactive image segmentation, where each pixel is assigned either to foreground (FG) or background (BG). The segmentation problem is formulated as the minimization of an energy function of the form:

$$E(\mathbf{x}) = \sum_p D_p(x_p) + \sum_{(p,q) \in \mathcal{N}} V_{pq}(x_p, x_q) \quad (1)$$

where $x_p \in \{\text{FG}, \text{BG}\}$ denotes the label of pixel p . The unary term D_p encodes the likelihood of a pixel belonging to foreground or background, while the pairwise term V_{pq} enforces spatial smoothness between neighboring pixels.

2.2 Unary Terms: Color Modeling

The unary costs are derived from color models learned from user-provided scribbles. Foreground and background scribbles are used as hard constraints and as training data for the color models.

We model the color distribution in CIELAB space using Gaussian Mixture Models (GMMs). For each pixel p , the unary costs are defined as the negative log-likelihoods:

$$D_p(\text{FG}) = -\log P(I_p \mid \text{GMM}_{\text{FG}}) \quad (2)$$

$$D_p(\text{BG}) = -\log P(I_p \mid \text{GMM}_{\text{BG}}) \quad (3)$$

To ensure numerical stability and non-negative costs, both unary terms are shifted by their minimum value per pixel. Scribble pixels are enforced by assigning infinite cost to the opposite label.

Compared to simple color histograms, GMMs provide a smoother and more expressive representation of multimodal color distributions, which is especially beneficial for objects with varying appearance.

2.3 Pairwise Terms: Spatial Smoothness

To encourage spatial coherence of the segmentation, we use a contrast-sensitive Potts model as pairwise term. For neighboring pixels (p, q) , the pairwise cost is defined as:

$$V_{pq}(x_p, x_q) = \begin{cases} 0, & x_p = x_q \\ \lambda \exp\left(-\frac{\|I_p^{ab} - I_q^{ab}\|_1}{\sigma}\right), & x_p \neq x_q \end{cases} \quad (4)$$

where I^{ab} denotes the chromatic channels of the LAB color space. This formulation penalizes label changes more strongly in homogeneous regions and allows discontinuities at strong color boundaries.

An 8-connected neighborhood system is used to capture diagonal structures in the image.

We tuned λ and σ over several values and obtained the best results with $\lambda = \sigma = 10$.

2.4 Optimization via s-t Min-Cut

The energy minimization problem is solved exactly using an s-t min-cut / max-flow algorithm. Each pixel corresponds to a node in the graph, with edges to the source (foreground) and sink (background) encoding the unary costs. Pairwise terms are modeled as undirected edges between neighboring pixels.

The global minimum of the binary energy is obtained by computing the minimum cut of the graph. We use the Boykov-Kolmogorov max-flow algorithm as provided by the `PyMaxflow` library.

To improve robustness, we perform an iterative refinement procedure similar to GrabCut. After each cut, the foreground and background GMMs are re-estimated using the current segmentation result, while keeping the original scribble pixels fixed. To prevent model drift, only a morphological core of the foreground region is used for re-training.

In all experiments, we ran the iterative refinement for 10 iterations, which provided stable convergence in practice.

This iterative scheme significantly improves segmentation quality for objects with heterogeneous appearance.

2.5 Experimental Results and IoU Evaluation

The segmentation results are evaluated using the Intersection over Union (IoU) metric:

$$\text{IoU} = \frac{|\text{Prediction} \cap \text{Ground Truth}|}{|\text{Prediction} \cup \text{Ground Truth}|} \quad (5)$$

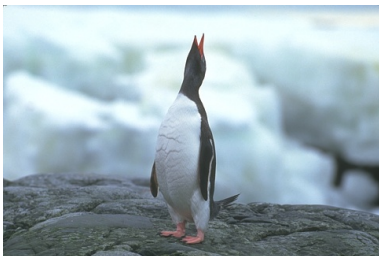
The evaluation is performed on six images from the dataset. For each image, user-provided foreground and background scribbles are used to learn the color models, and the resulting segmentation is compared against the corresponding ground truth annotation. Figure 1 shows all original input images used in the experiments. The average IoU across the evaluated images exceeds the required threshold of 0.50. Qualitative results demonstrate that the proposed method is able to segment the main object reliably even with sparse scribble annotations.

2.6 Discussion

We observe that GMM-based color modeling consistently outperforms simple histograms, especially for objects with complex or multimodal color distributions. Histogram models tend to be overly sensitive to sparse scribbles, while GMMs generalize better.

The choice of pairwise parameters λ and σ is crucial: too weak smoothness leads to noisy segmentations, while overly strong smoothness may cause leakage into the background.

Iterative re-estimation of the color models significantly improves results but requires careful handling to avoid model drift. Enforcing scribble constraints and restricting re-training to a foreground core proved essential for stable convergence.



(a) 106024



(b) 208001



(c) aero_2008_002358



(d) bike_2007_005878



(e) person7



(f) scissors

Figure 1: Input images used for evaluation (2 rows $\times$ 3 columns).

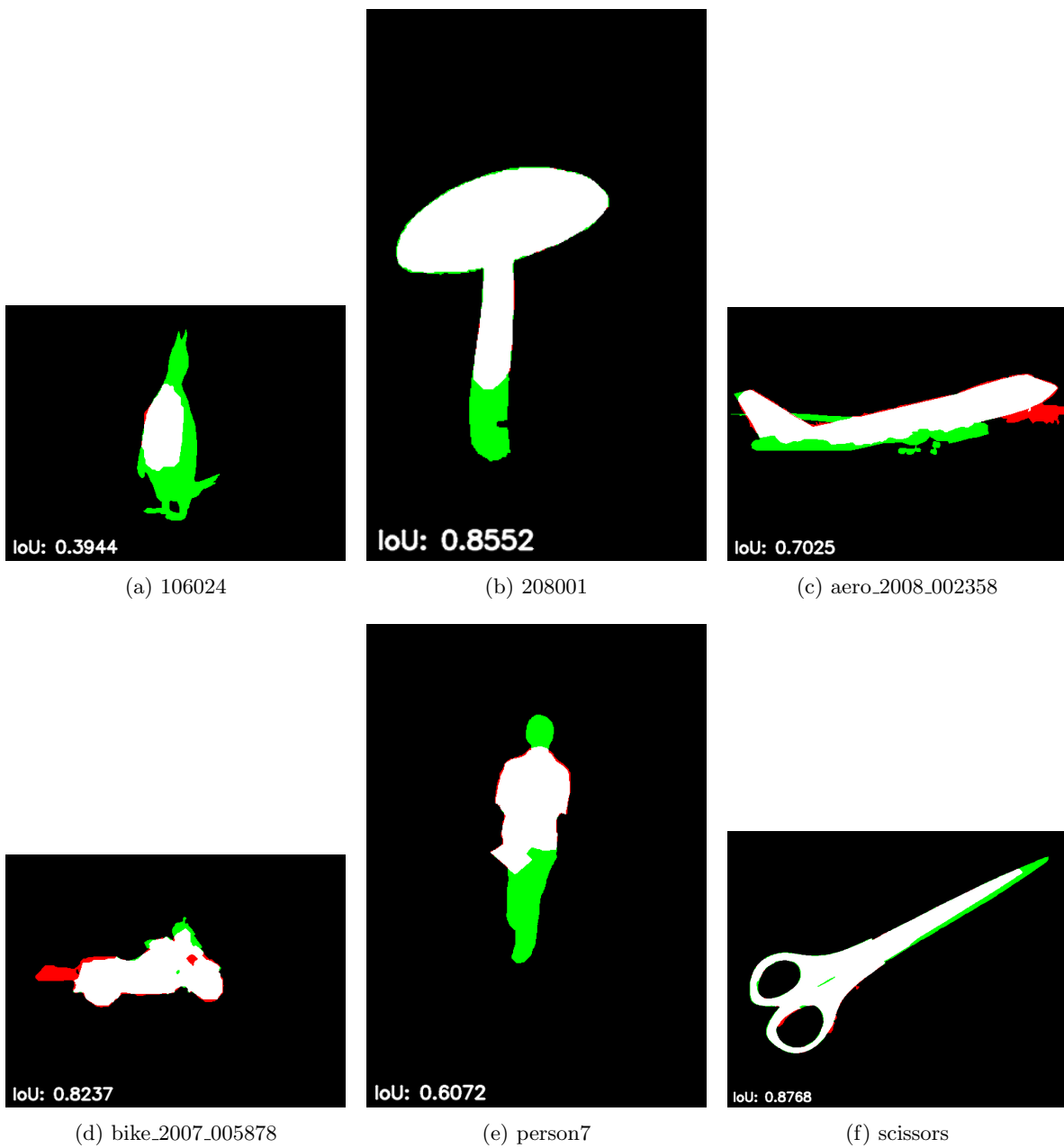


Figure 2: IoU overlay visualizations for all evaluated images. Ground truth foreground is shown in green, the predicted foreground segmentation in red. White regions indicate overlap between prediction and ground truth, while black corresponds to background.